LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 4
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1. TITLE

Develop a Dynamic Webpage Using HTML5, CSS3, and JavaScript

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2. AIM

To design and develop a dynamic webpage using HTML5, CSS3, and JavaScript that implements a Counter App, demonstrating DOM manipulation, external file linking, event handling via onclick attributes, and CSS Flexbox-based layout and button styling.

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3. OBJECTIVE

The objectives of this experiment are:
- To understand the structure of a multi-file web project with separate HTML, CSS, and JavaScript files.
- To link an external CSS stylesheet and an external JavaScript file to an HTML document.
- To use the element with an id to display a dynamically updated value on the page.
- To implement onclick event handlers on buttons to call JavaScript functions.
- To manipulate the DOM using document.getElementById() and .innerText to update the counter display.
- To use CSS Flexbox properties to centre page content both horizontally and vertically.
- To apply button styling including background colour, border-radius, padding, and hover effects using CSS pseudo-classes.

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4. THEORY

Dynamic Web Page — a web page whose content changes in response to user interactions without reloading, achieved by using JavaScript to read and modify HTML elements through the Document Object Model (DOM).

DOM Manipulation — the process of accessing and modifying HTML elements from JavaScript at runtime. document.getElementById(id) returns the element with the matching id attribute, and properties such as innerText allow its content to be read or updated.

Event Handling — onclick is an HTML attribute that binds a JavaScript function call to a button click event. When the button is clicked, the browser calls the named function defined in the linked script file.

External Files — separating HTML structure, CSS presentation, and JavaScript behaviour into distinct files improves maintainability. The HTML links them using <link rel="stylesheet"> for CSS and <script src="..."> for JavaScript.

CSS Flexbox — a layout model that aligns and distributes space among items in a container. Setting display: flex with justify-content: center and align-items: center on the body centres the main container in the viewport.

CSS Pseudo-classes — selectors that apply styles based on element state. The button:hover rule changes the button background when the pointer is over it, providing interactive visual feedback without JavaScript.

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5. ALGORITHM / PROCEDURE

- Create three files in the same folder: Index.html, style.css, and script.js.
- In Index.html, add the DOCTYPE declaration, <html>, <head>, and <body> elements.
- Inside <head>, link style.css using <link rel="stylesheet" href="style.css">.
- In <body>, add a containing an <h1> heading and a nested for the Counter App.
- Inside the Counter App , add an <h2>, a <p> with a 0 to display the count, and three <button> elements with onclick attributes calling increaseCount(), decreaseCount(), and resetCount().
- Add <script src="script.js"></script> just before the closing of the main container.
- In style.css, set body to display: flex with justify-content: center and align-items: center to centre the layout.
- Style .main-container with width: 60% and margin: 0 auto, and h1 with a dark colour.
- Style the button selector with background-color, color, border, padding, margin, cursor, and border-radius properties.
- Add a button:hover rule to darken the background colour on mouse-over.
- In script.js, declare let count = 0 as the counter variable.
- Define increaseCount(): increment count by 1 and update the span innerText.
- Define decreaseCount(): decrement count by 1 and update the span innerText.
- Define resetCount(): set count to 0 and update the span innerText.
- Open Index.html in a browser, test all three buttons, and record expected vs. actual output for each test case.

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6. PROGRAM

Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My First Dynamic Webpage</title>
    <!-- Linking CSS file -->
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <div class="main-container">
        <h1>Dynamic Web Page</h1>
        <!-- Dynamic Counter -->
        <div>
            <h2>Counter App</h2>
            <p>Current Count: <span id="count-number">0</span></p>
            <button onclick="increaseCount()">Increase (+1)</button>
            <button onclick="decreaseCount()">Decrease (-1)</button>
            <button onclick="resetCount()">Reset</button>
        </div>
        <script src="script.js"></script>
    </div>
</body>
</html>
```

style.css

```
h1 {
    color: #333;
}
.main-container {
    width: 60%;
    margin: 0 auto;
}
```

```
body {
    font-family: Arial, sans-serif;
    background-color: #f0f8ff;
    display: flex;
    justify-content: center;
    align-items: center;
    min-height: 100vh;
    margin: 0;
    text-align: center;
}
button {
    background-color: #008CBA;
    color: white;
    border: none;
    padding: 8px 15px;
    margin: 5px;
    cursor: pointer;
    border-radius: 4px;
}
button:hover {
    background-color: #005f73;
}
```

script.js

```
// Variable for counter
let count = 0;

// 1. Counter Functions using document.getElementById
function increaseCount() {
    count = count + 1;
    document.getElementById("count-number").innerText = count;
}

function decreaseCount() {
    count = count - 1;
```

```
    document.getElementById("count-number").innerText = count;
}

function resetCount() {
    count = 0;
    document.getElementById("count-number").innerText = count;
}
```

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7. CODE EXPLANATION

Code	Explanation
<link rel="stylesheet" href="style.css">	Links the external CSS file style.css to the HTML page so all styling rules are applied.
<script src="script.js"></script>	Links the external JavaScript file script.js, placed just before the closing so the DOM elements are available when the script runs.
0	A span element that displays the current counter value. Its id allows JavaScript to target and update it directly.
onclick="increaseCount()"	An inline event handler that calls increaseCount() in script.js each time the Increase button is clicked.
let count = 0	Declares and initialises the counter variable to 0. This variable persists across function calls and tracks the current count.
count = count + 1	Increments the counter by 1 inside increaseCount(). The updated value is then written back to the DOM.
count = count - 1	Decrements the counter by 1 inside decreaseCount(). The updated value is then written back to the DOM.
count = 0 (resetCount)	Resets the counter variable to 0 inside resetCount(), returning the display to its initial state.
document.getElementById("count-number")	DOM method that locates the element with id="count-number" so its text content can be updated.
.innerText = count	Sets the visible text of the span element to the current value of count, updating the display immediately.
display: flex; justify-content: center	CSS Flexbox properties that centre the main container both horizontally and vertically within the viewport.
button:hover { background-color: #005f73; }	CSS pseudo-class rule that darkens the button background when the pointer hovers over it, providing visual feedback.

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8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Load Index.html in a browser	Page renders with heading "Dynamic Web Page", "Counter App" subheading, count showing 0, and three buttons	Page rendered correctly with all elements visible	Pass
TC-02	Click Increase (+1) once	Count display updates from 0 to 1	Count displayed as 1	Pass
TC-03	Click Increase (+1) five times	Count display updates to 5	Count displayed as 5	Pass
TC-04	Click Decrease (-1) once	Count display decrements by 1	Count decremented correctly	Pass
TC-05	Click Decrease (-1) when count is 0	Count display updates to -1 (no lower bound enforced)	Count displayed as -1	Pass
TC-06	Click Reset after several increments	Count display resets to 0	Count displayed as 0	Pass
TC-07	Hover over a button	Button background changes to darker colour (#005f73)	Hover colour applied correctly	Pass
TC-08	Inspect page layout	Main container centred horizontally and vertically on the page with light blue background	Flexbox centering applied correctly	Pass
TC-09	Inspect button styling	Buttons display with blue background, white text, rounded corners, and spacing	Button styles rendered as defined in style.css	Pass
TC-10	Alternate Increase and Decrease clicks	Count value reflects the net result of all clicks correctly	Counter updated accurately for each click	Pass

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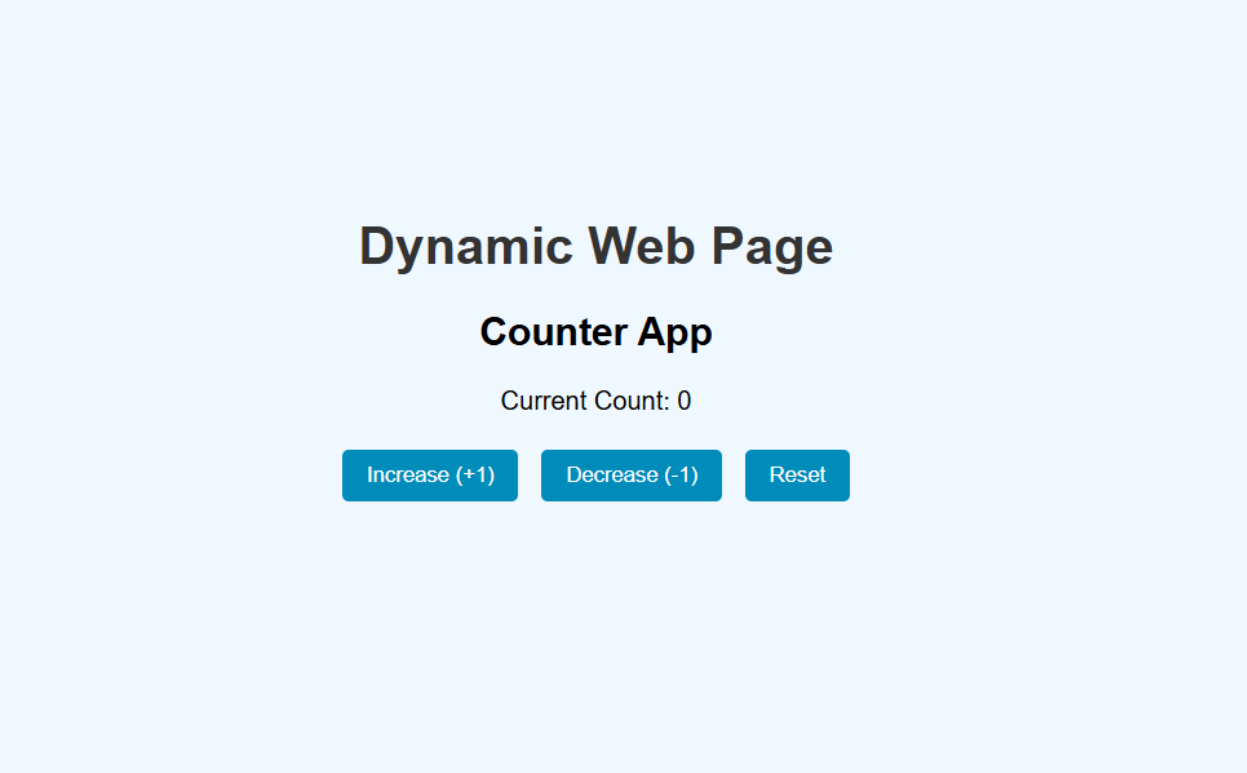
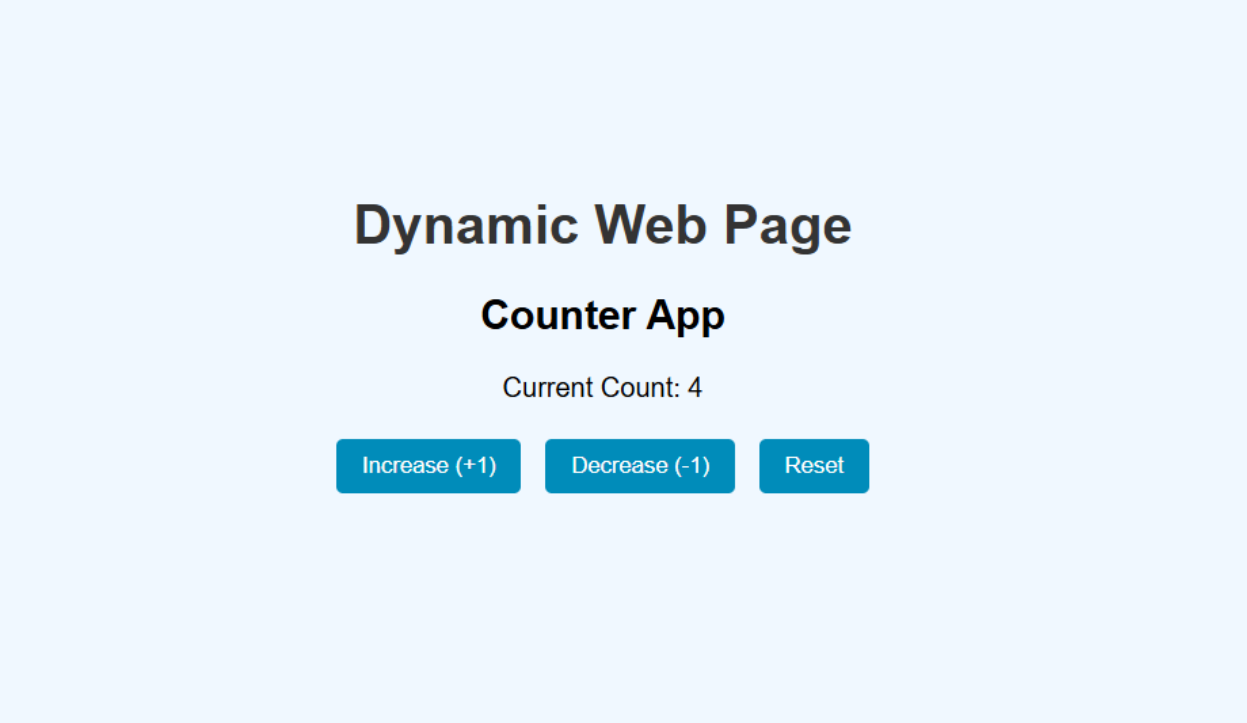
9. OUTPUT

Output 1: Initial Page Load

The browser displays the "Dynamic Web Page" heading centred on a light blue background. The Counter App section shows "Current Count: 0" and three styled blue buttons — Increase (+1), Decrease (-1), and



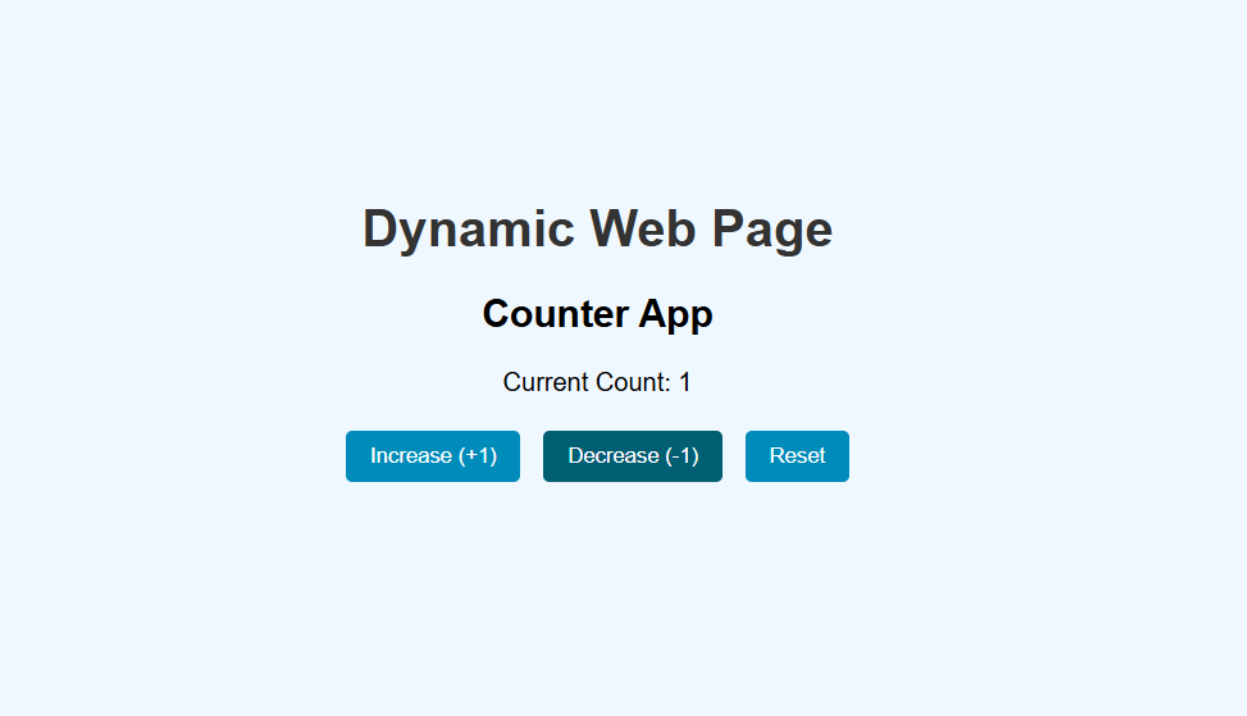
Output 2: Increase Button



Each click on the Increase (+1) button increments the displayed count by 1. The span element updates immediately without a page reload, demonstrating live DOM manipulation via innerText.

Output 3: Decrease Button

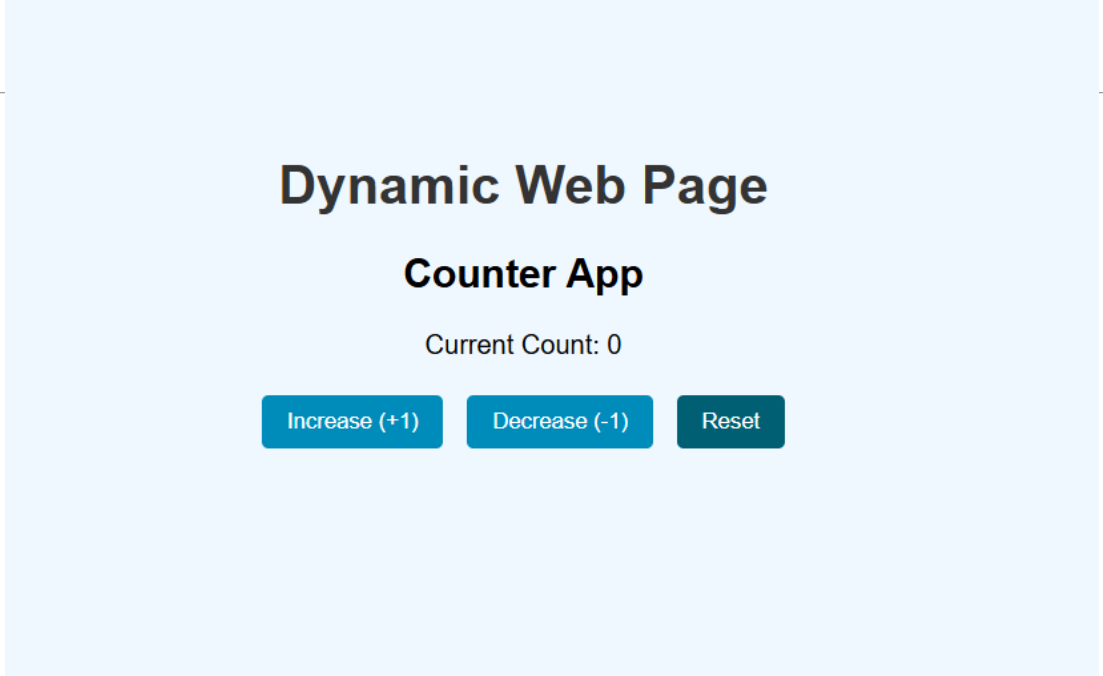
Clicking Decrease (-1) decrements the count by 1 on each click, including below zero and updates the display.



Output 4: Reset Button

Clicking Reset sets the count back to 0 instantly, confirming that resetCount() correctly reassigns the variable and updates the display.

10.



RESULT

Thus, a dynamic webpage was successfully designed and developed using HTML5, CSS3, and JavaScript across three linked files. The page demonstrates external file linking, DOM manipulation using document.getElementById() and innerText, onclick event handling, CSS Flexbox-based centred layout, and interactive button styling with hover effects. All ten test cases passed successfully, confirming correct counter behaviour, layout, and styling.